DIGITAL FORENSICS INVESTIGATION REPORT

Lab 2 — Data Carving and Raw File Recovery

Case Reference	ICDFA-CIP-B101-Lab02
Evidence Files	Ch01InChap01.dd	J_ub_law.jpg	120M.7z
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Classification	Training — ICDFA Lab Submission

ICDFA Fellowship — Web Application Security and Digital Forensics

1. Executive Summary

This report documents the digital forensic investigation conducted as part of the ICDFA CIP-B101 Lab 2 exercise on Data Carving and Raw File Recovery. The investigation examined three evidence files: a forensic disk image (Ch01InChap01.dd), a JPEG sample for hex and signature analysis (J_ub_law.jpg), and a USB forensic image (120M.7z).

The investigation covered file signature identification using xxd, metadata extraction using strings, embedded file discovery using Binwalk, file system analysis using The Sleuth Kit, and USB image preparation for Scalpel file carving. All evidence was hashed before analysis to verify integrity. The original evidence files were not modified at any point during the investigation.

Note: Part F (Scalpel file carving) was partially completed — configuration was initiated but the full carving operation will be completed and documented as a supplementary submission.

2. Evidence Details

Evidence File	Size	MD5 Hash	SHA-256 Hash
Ch01InChap01.dd	1.5MB	a117773bcf1fc88ec0ab8e0a349fbb[illegible]	3de8053e4f3d9c8ab98b3aadb2480685...
J_ub_law.jpg	1.7MB	83a360ac7f7e0ca318e5bfe39f95f1[illegible]	238ff34393c50e52c0e8b14fcff8ec7d...
120M.7z	36MB	dfe7b5424e54cd1bf50d5df47aceeb[illegible]	2d2a3d93c9ec65bcad9f89c9894429ea...
usb_fat_carving.001	119MB	ba4a1d0ba49f4a6667b00a3b3e85e604	Verified by FTK Imager report

3. Part A — File Signatures and Hex Analysis

3.1 JPEG Header Identification

The xxd tool was used to examine the raw binary content of J_ub_law.jpg. xxd converts binary file content into hexadecimal representation, allowing forensic analysts to identify file signatures, hidden content and file structure.

Command:

```
xxd J_ub_law.jpg | head -20
```

Key findings from header analysis:

- First two bytes: FF D8 — JPEG Start of Image (SOI) signature
- Bytes 3-4: FF E1 — EXIF data marker
- Camera: NIKON CORPORATION — visible in plain text at offset 0x00e0
- Model: NIKON D4 — identified at offset 0x00f0
- Software: Adobe Photoshop 21.1 (Macintosh) — at offset 0x0110
- Date: 2020:08:20 10:20:05 — at offset 0x0120

3.2 JPEG Footer Identification

Command:

```
xxd J_ub_law.jpg | tail -20
```

The last two bytes of the file were found to be FF D9 — the JPEG End of Image (EOI) signature. This confirms the file is a complete, valid JPEG image.

Signature	Hex Value	Location	Meaning
JPEG SOI (Header)	FF D8	Offset 0x00000000	Start of Image — file begins here
JPEG EOI (Footer)	FF D9	Offset 0x0019d1f6	End of Image — file ends here

3.3 Hex Dump and Reconstruction

Commands:

```
xxd -p J_ub_law.jpg > PartA_hexdump.txt
```

```
xxd -r -p PartA_hexdump.txt > J_ub_law_reconstructed.jpg
```

A plain hexadecimal dump of J_ub_law.jpg was created (3.3MB text file). The image was then reconstructed from this hex dump using the reverse xxd function. Hash comparison confirmed the reconstruction was byte-perfect:

File	MD5 Hash	Result
J_ub_law.jpg (Original)	83a360ac7f7e0ca318e5bfe39f95f137	MATCH
J_ub_law_reconstructed.jpg	83a360ac7f7e0ca318e5bfe39f95f137	MATCH

4. Part B — Embedded Metadata Analysis

The strings command was used to extract all human-readable text embedded inside J_ub_law.jpg. This reveals EXIF metadata without requiring a dedicated metadata tool.

Command:

```
strings J_ub_law.jpg | head -50
```

Metadata Field	Value Found	Forensic Significance
Camera Manufacturer	NIKON CORPORATION	Identifies the device brand
Camera Model	NIKON D4	Professional DSLR camera
Editing Software	Adobe Photoshop 21.1	Image was post-processed
Operating System	Macintosh	Edited on Apple Mac computer
Date Photo Taken	2020:08:20 10:20:05	Exact date and time confirmed
Photographer Name	HOWARD KORN	Person of interest identified
Lens Used	17.0-35.0 mm f/2.8	Professional lens equipment
Camera Original Date	2013:10:08 18:56:21	Camera manufacture/setup date

Forensic Conclusion: The photographer is identified as HOWARD KORN. The photo was taken on 20 August 2020 using a Nikon D4 professional DSLR camera with a 17-35mm f/2.8 lens and subsequently edited using Adobe Photoshop 21.1 on a Macintosh computer. This level of detail would be significant in a real investigation.

4.1 Personal Photo Metadata Analysis

As an additional practical exercise, the same strings analysis was applied to a personal JPEG photo (yomide.jpeg, 117KB). The analysis returned no identifiable metadata — no camera model, date, GPS or author information was found.

Conclusion: The photo had its metadata stripped prior to receipt, consistent with sharing via messaging applications such as WhatsApp or Telegram. These platforms compress images and remove EXIF metadata automatically. This demonstrates why forensic analysts cannot always rely on metadata being present in recovered images and must seek the original source device.

5. Part C — Sleuth Kit Analysis

The Sleuth Kit command-line tools were used to perform a structured examination of Ch01InChap01.dd, progressing from image-level analysis down to individual file recovery.

Step	Command	Purpose	Key Finding
1	`img_stat`	Image type and size	Raw image, 1,474,560 bytes, 512-byte sectors
2	`mmls`	Partition layout	No partition table — single FAT12 file system, offset 0
3	`fsstat`	File system details	FAT12, data area at sector 19, cluster size 512
4	`fls`	List all files	6 files found — 4 deleted (marked with *)
5	`istat 13`	INCOME.XLS metadata	13,824 bytes, created 2005-12-09, sectors 285-311
6	`icat 13`	Recover INCOME.XLS	14K file recovered successfully
7	`blkcat 285`	Raw block extraction	Sector 285 extracted — 512 bytes (first sector of file)

5.1 File Listing Summary

Inode	File Name	Status	Notes
5	Client Info.mdb	Active	Microsoft Access database
8	Billing Letter.doc	DELETED	Recovered by tsk_recover in Lab 1
11	confirmation.txt	DELETED	FTP credentials found inside
13	Income.xls	Active	Target file — recovered using icat
15	letter1.txt	DELETED	Meeting request from George
17	Regrets.doc	DELETED	Recovered by tsk_recover in Lab 1

6. Part D — Binwalk Embedded File Analysis

Binwalk was used to scan evidence files for embedded file structures or signatures. Binwalk identifies hidden files embedded inside other files by scanning raw bytes for known file signatures.

6.1 Binwalk on J_ub_law.jpg

Command:

```
binwalk J_ub_law.jpg
```

Decimal Offset	Hex Offset	Description
0	0x0	JPEG image data, EXIF standard
12	0xC	TIFF image data (embedded thumbnail in EXIF)
26844	0x68DC	Adobe copyright string — Copyright 1999 Adobe Systems

Analysis: No hidden or suspicious files were detected. The TIFF data at offset 12 is a standard thumbnail embedded in EXIF metadata. The Adobe copyright string confirms image processing with Adobe software. The image is forensically clean.

6.2 Binwalk on Ch01InChap01.dd

Command:

```
binwalk Ch01InChap01.dd
```

Output: No embedded file signatures detected. This is expected for a FAT12 floppy disk image — files are accessed through the file system rather than embedded at the binary level. Sleuth Kit tools are the appropriate method for examining this image type.

6.3 Limitation — File_carving.docx

The instructor-provided File_carving.docx was not available for this exercise. The Binwalk manual and automatic extraction workflow was therefore demonstrated using the available evidence files. This limitation is noted for completeness.

7. Part E — USB Image Preparation

7.1 Extraction of 120M.7z

Command:

```
7z x 120M.7z
```

The compressed archive was extracted using 7zip, revealing a folder containing two files: usb_fat_carving.001 (119MB USB forensic image) and usb_fat_carving.001.txt (FTK Imager acquisition report).

7.2 FTK Imager Acquisition Report

Field	Value
Acquisition Tool	AccessData FTK Imager 4.5.0.3
Case Number	USB File Carving
Evidence Number	UB-002
Examiner	Frank Xu
Institution	University of Baltimore
Source Device	General UDisk USB Device
Source Size	119MB — 243,712 sectors
Acquisition Date	Wednesday 22 September 2021
MD5 (FTK Verified)	ba4a1d0ba49f4a6667b00a3b3e85e604 — VERIFIED
SHA1 (FTK Verified)	bcc2d49fd49c9521ecb1739f6542c6bf327375ef — VERIFIED

7.3 Independent Hash Verification

Command:

```
md5sum usb_fat_carving.001
```

Result: ba4a1d0ba49f4a6667b00a3b3e85e604 — MATCHES FTK report. Evidence integrity confirmed.

7.4 Partition Analysis — mmls

Command:

```
mmls usb_fat_carving.001
```

Slot	Start Sector	End Sector	Description
Meta	0	0	Primary Partition Table
Unallocated	0	127	128 sectors before partition
000:000	128	243711	Win95 FAT16 (0x0e) — Main partition

Partition offset confirmed as 128 sectors. This matches the lab reference value. The file system is FAT16 — commonly used on USB drives.

7.5 File System Analysis — fsstat

Command:

```
fsstat -o 128 usb_fat_carving.001
```

- File System Type: FAT16

- Volume Label: USB
- Sectors before file system: 128 (offset confirmed)
- Data Area: starts at sector 480
- Sector Size: 512 bytes
- Cluster Size: 2048 bytes

7.6 File Listing — fls

Command:

```
fls -o 128 usb_fat_carving.001
```

35 files were found on the USB image. 32 were deleted (marked with *). Active files: readme.docx, readme.txt, .dropbox.device.

File Type	Deleted Files Found
JPEG Images	J_ub_law.jpg, J_ub_night.jpg
BMP Images	B_ub_poe4.bmp, B_zoom-eubie-mono.bmp
GIF Images	G_BuiltForThis.gif, G_zoom-sc.gif
PDF Documents	pd_Evidence_search_techniques.pdf, pd_Forensic_Report_Template.pdf
Word Documents	DO_example.doc, DO_example2.doc, Chinese-language .doc
PowerPoint	pp_Number_Systems.pptx, pp_one_page.pptx
Other	Java files, HTML files, TIFF, WAV, ZIP, PST, RTF, PNG

8. Part F — Scalpel File Carving

Scalpel is a file carving tool that recovers deleted files by scanning raw disk images for known file signatures (headers and footers). Unlike file system-based recovery tools, Scalpel works even when file system metadata is missing or corrupt.

8.1 Scalpel Configuration

Command:

```
sudo nano /etc/scalpel/scalpel.conf
```

The Scalpel configuration file was opened. JPEG file type entries were identified and uncommented to enable JPEG carving. The relevant lines in the configuration are:

```
jpg y 5242880 \xff\xd8\xff???Exif \xff\xd9 REVERSE
```

```
jpg y 5242880 \xff\xd8\xff???JFIF \xff\xd9 REVERSE
```

8.2 Limitation — Scalpel Carving Not Completed

The full Scalpel carving operation against usb_fat_carving.001 was not completed during this lab session due to time constraints and VM performance limitations. The configuration was initiated and the JPEG file type entries were identified. The carving operation will be completed and documented as a supplementary submission. Expected output would include recovered JPEG files, an audit.txt file, and hash verification of recovered evidence.

9. Conclusion

This laboratory exercise successfully demonstrated core data carving and raw file recovery concepts. File signatures were identified using xxd, confirming JPEG files begin with FF D8 and end with FF D9. A JPEG image was successfully reconstructed from its hexadecimal dump with hash verification confirming byte-perfect reconstruction.

Metadata analysis using strings revealed significant forensic information from J_ub_law.jpg including the photographer's name (HOWARD KORN), camera model (Nikon D4), editing software and exact date of capture. Binwalk confirmed no hidden files were embedded in the evidence.

Sleuth Kit analysis of Ch01InChap01.dd identified a FAT12 file system containing 6 files, 4 of which were deleted. INCOME.XLS was successfully recovered using icat. The USB image (usb_fat_carving.001) was prepared for carving with partition offset 128 confirmed, 32 deleted files identified across multiple file types.

All evidence hashes were verified before and after analysis. The original evidence files were not modified at any point. All commands, outputs and recovered files are documented in the accompanying GitHub repository: https://github.com/Adeleye001/ICDFA-Ch02-Forensics-Lab

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